

File Transfer Services Using SFTP and VSFTPD

Objective

GOAL: Enable file transaction to servers using file transfer protocol [FTP]

1. Configure SFTP in an instance such that it can receive files via port number **22** to a specific directory as a specific user. The SFTP users should have restricted privileges.
2. Verify the working with an FTP client such as **FileZilla** or **Cyberduck**.
3. Install and configure **VSFTPD** in an instance such that it can receive files securely via port **21** to the configured directory as a configured user. The VSFTPD user should have restricted privileges.
4. Verify the working with an FTP client.

1. SFTP Configuration

1.1 Create the SFTP User

A dedicated user named `sftpuser` was created for SFTP file transfers. A separate user was used to ensure that file-transfer access remains isolated from normal system users.

1.2 Create the Restricted Directory Structure

The following directory structure was created:

```
/sftp
├── sftpuser
│   └── files
```

The `/sftp` and `/sftp/sftpuser` directories were owned by `root`, while the `files` directory was owned by `sftpuser`.

This prevents the SFTP user from modifying the `chroot` directory while still allowing file uploads inside the designated `files` directory.

1.3 Configure SSH for Restricted SFTP

The SSH configuration was modified to use the internal SFTP subsystem:

```
Match User sftpuser
    ForceCommand internal-sftp
    ChrootDirectory /sftp/sftpuser
    X11Forwarding no
    AllowTcpForwarding no
```

The configuration restricts sftpuser to the /sftp/sftpuser environment and prevents access to the normal server filesystem.

1.4 SFTP Security

The SFTP user is restricted using:

- **Host:** 3.109.158.151
- **Port:** 22
- **Authentication:** User authentication
- **Chroot:** /sftp/sftpuser
- **SFTP subsystem:** Internal SFTP
- **Port forwarding:** Disabled
- **X11 forwarding:** Disabled
- **Writable directory:** /files

```
[ec2-user@ip-172-31-10-10 ~]$ sftp sftpuser@localhost
sftpuser@localhost's password:
Connected to localhost.
sftp> pwd
Remote working directory: /
sftp> cd files
sftp> cd ..
sftp> cd /etc
stat remote: No such file or directory
sftp> cd files
sftp> ls
sftp-test.txt          test file sftp.txt
sftp> |
```

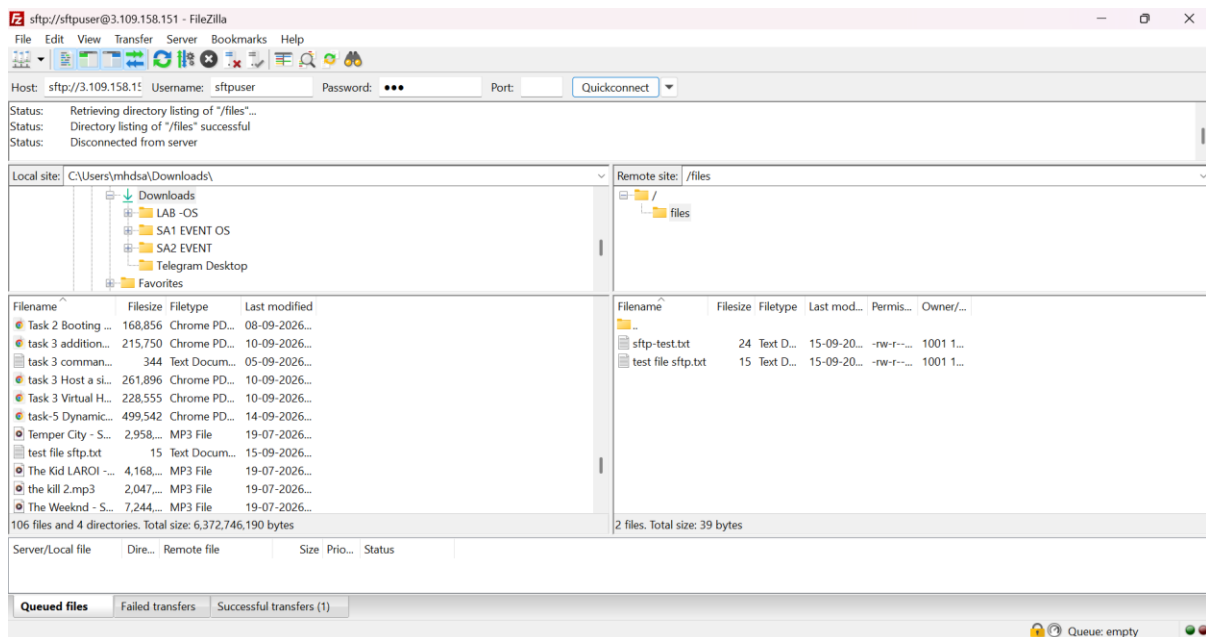
2. SFTP Verification Using FileZilla

FileZilla was configured using:

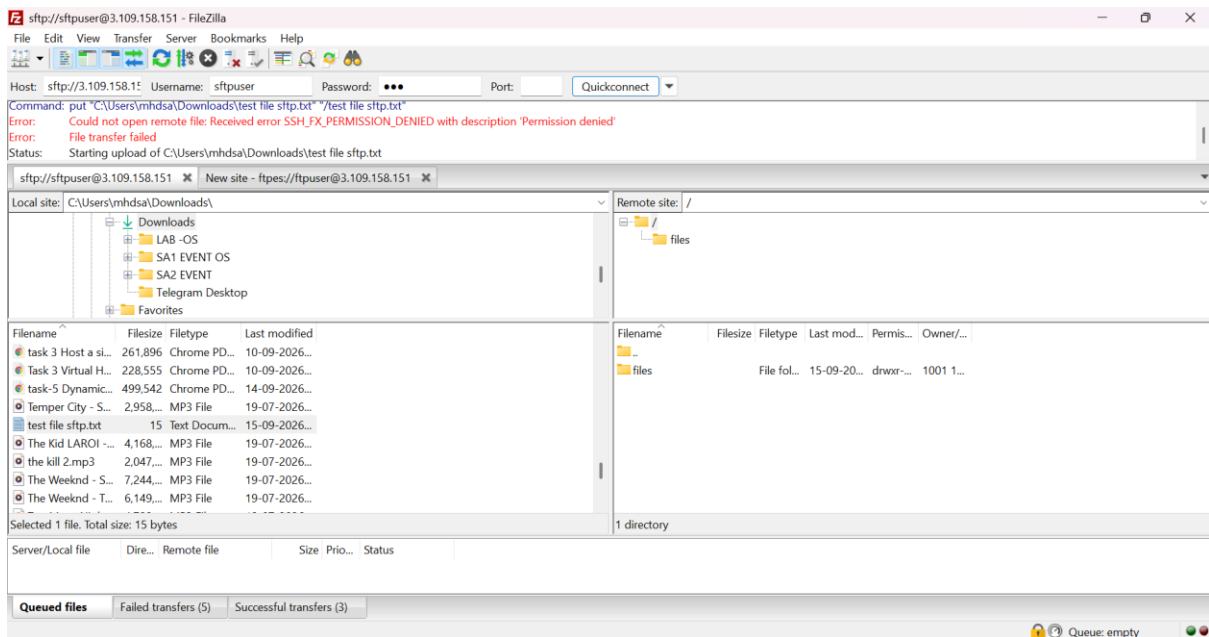
Setting	Value
Protocol	SFTP - SSH File Transfer Protocol
Host	3.109.158.151
Port	22
User	sftpuser
Logon Type	Normal

The connection was successfully established and the /files directory was accessible.

A test file was uploaded successfully to the /files directory.



The restriction was also verified by attempting to upload a file outside the permitted writable directory. The operation was denied with **Permission denied**, confirming that the SFTP user's privileges are restricted.



3. VSFTPD Installation and Configuration

3.1 Install VSFTPD

The **VSFTPD** package was installed on the Linux server.

VSFTPD provides FTP functionality, while **TLS encryption** was configured to secure the FTP connection.

3.2 Create the FTP User

A dedicated FTP user named:

`ftpuser`

was created for file transfers.

3.3 Create the FTP Directory Structure

The following directory structure was created:

```
/ftp
├── ftpuser
│   └── files
```

The directory ownership was configured as:

```
/ftp          → root:root
/ftp/ftpuser   → root:root
/ftp/ftpuser/files → ftpuser:ftpuser
```

This prevents ftpuser from writing to the FTP root while allowing file uploads inside the files directory.

4. Secure VSFTPD Configuration

VSFTPD was configured to disable anonymous access and allow authenticated local users.

The important security settings include:

```
anonymous_enable=NO
local_enable=YES
write_enable=YES

chroot_local_user=YES
local_root=/ftp/ftpuser
```

TLS Encryption

Because plain FTP transmits credentials and data without encryption, **FTP over TLS (FTPS)** was configured.

The following settings were enabled:

```
ssl_enable=YES
force_local_logins_ssl=YES
force_local_data_ssl=YES

rsa_cert_file=/etc/pki/tls/certs/vsftpd.crt
rsa_private_key_file=/etc/pki/tls/private/vsftpd.key
```

This ensures that:

- User credentials are transmitted through an encrypted TLS connection.
- File-transfer data is encrypted.

- Anonymous login is disabled.
- The FTP user is restricted to its configured root directory.
- A TLS certificate and private key are used for secure communication.

Passive Mode Configuration

Passive FTP ports were configured to support external FTP clients:

```
pasv_enable=YES  
pasv_min_port=40000  
pasv_max_port=40010  
pasv_address=3.109.158.151
```

The corresponding passive port range **40000–40010** was allowed in the AWS Security Group.

5. SELinux Configuration

SELinux was kept **enabled and enforcing** rather than being disabled.

The FTP directories were given appropriate SELinux contexts.

The FTP root was labeled:

```
public_content_t
```

and the writable upload directory was labeled:

```
public_content_rw_t
```

The writable FTP content was therefore allowed through SELinux while maintaining the security of the system.

The appropriate SELinux setting was also enabled to permit FTP services to write to the designated writable content.

6. VSFTPD Verification Using FileZilla

FileZilla was configured with the following settings:

Setting	Value
Protocol	FTP - File Transfer Protocol
Host	3.109.158.151
Port	21
Encryption	Require explicit FTP over TLS
Logon Type	Normal
User	ftpusers

The connection was successfully established using **FTP over TLS**.

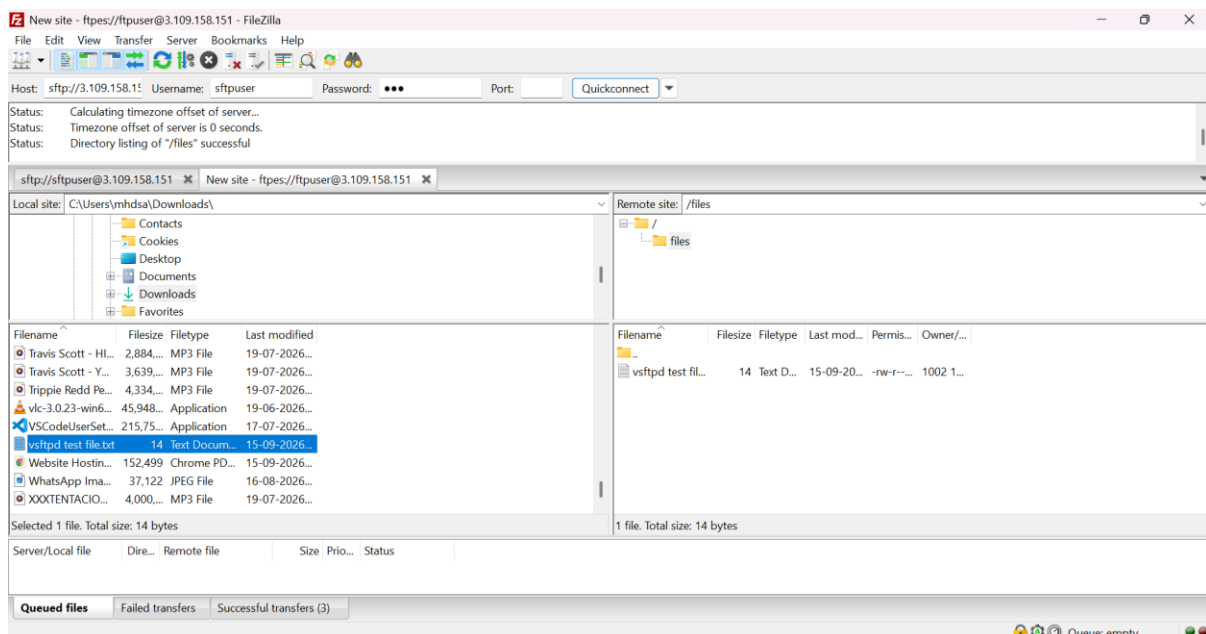
The remote FTP root was displayed as:

/

and the configured writable directory was available as:

/files

A test file was successfully uploaded to the /files directory.



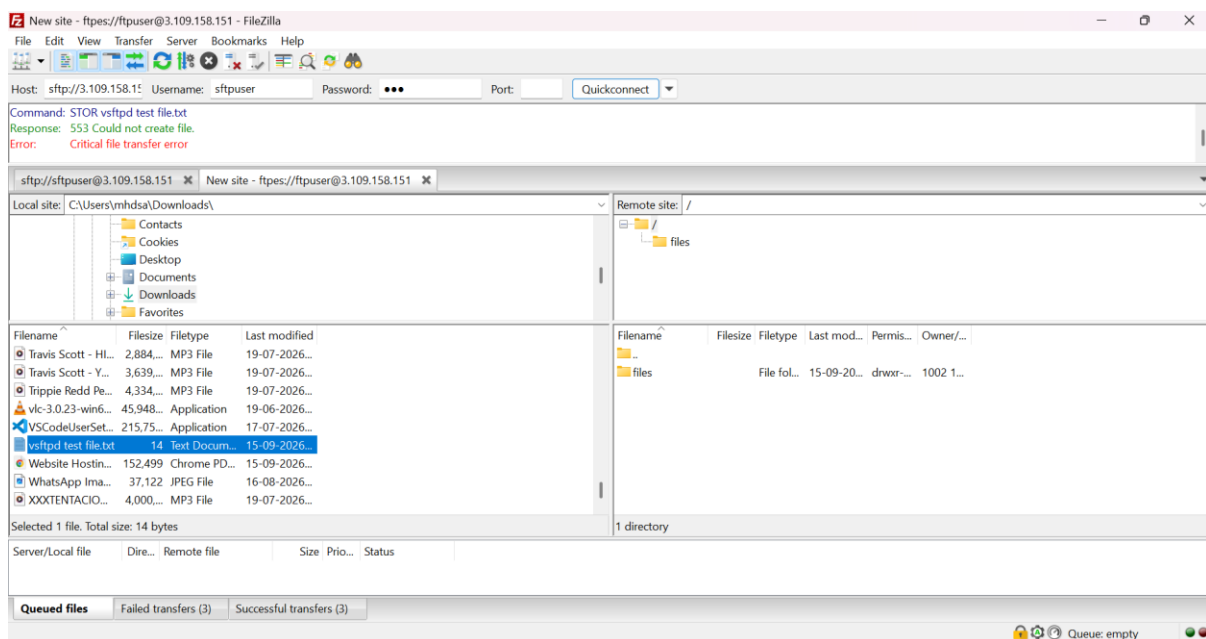
7. Restricted Privilege Verification

To verify the restriction applied to ftpuser, an attempt was made to upload a file directly to the FTP root /.

The server rejected the operation with:

553 Could not create file.

This confirms that the FTP user cannot write to the FTP root and can upload files only to the designated writable /files directory.



8. Security Features Implemented

Feature	Implementation
SFTP encryption	SSH
SFTP host	3.109.158.151
SFTP port	22
SFTP user restriction	Chroot
SFTP writable location	/files
Anonymous VSFTPD access	Disabled
Secure FTP	FTPS/TLS
FTPS host	3.109.158.151
FTPS port	21
Login encryption	Enabled

Data encryption	Enabled
VSFTPD user restriction	Chroot
VSFTPD writable location	/files
SELinux	Enforcing
SELinux FTP context	Configured
Passive FTP ports	40000–40010
AWS Security Group	Configured for required ports

Conclusion

SFTP and VSFTPD were successfully configured and tested for secure file transfer.

The **SFTP configuration** uses SSH encryption on port **22** and restricts sftpuser to its designated chroot environment.

The **VSFTPD configuration** uses **FTP over TLS (FTPS)** on port **21**, with encrypted authentication and file transfers. Anonymous access was disabled, and ftpuser was restricted to the configured FTP environment with /files as the writable directory.

SELinux was maintained in **Enforcing** mode and correctly configured for the FTP directories. FileZilla was successfully used to verify both secure file transfer and restricted user privileges.